

Day 50 - DHCP Snooping

Wednesday, September 9, 2026

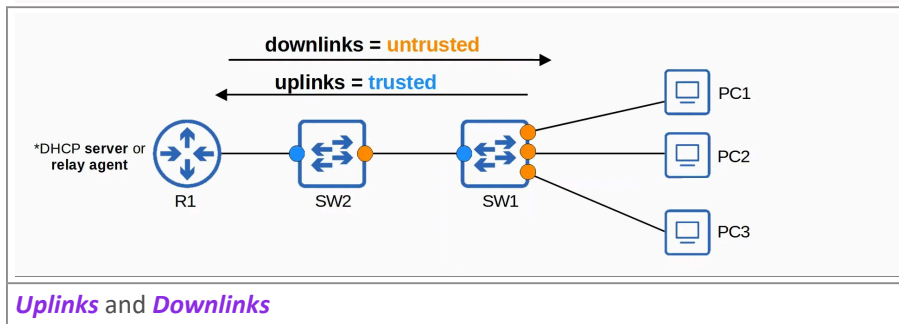
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DHCP Snooping Info

DHCP Snooping General Information

DHCP snooping is a security feature of switches that is used to filter DHCP messages received on *untrusted* ports. It only filters DHCP messages, anything else is not affected.

- By default, all ports are *untrusted*. There's also a standard to follow while assigning *trusted* ports.
 - Uplink* ports are configured as *trusted* ports. (The ports pointing towards network devices)
 - Downlink* ports are configured as *untrusted* ports. (The ports pointing towards end hosts)

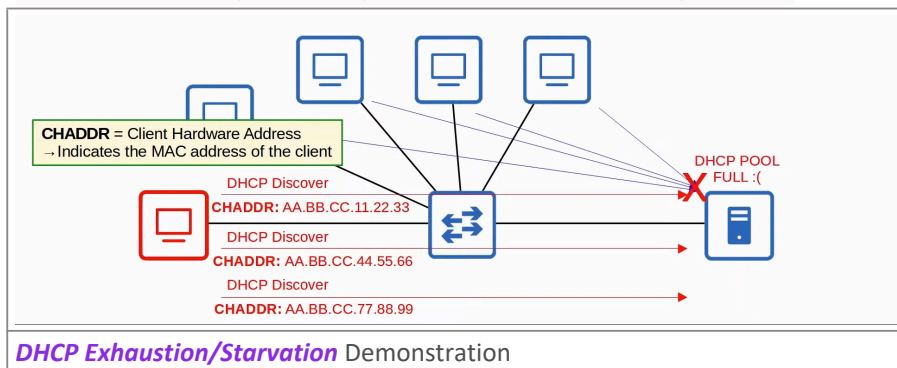


Example Attacks

1. DHCP Exhaustion/Starvation

An example of a DHCP-based attack is *DHCP Exhaustion/Starvation*.

- An *attacker* uses spoofed MAC addresses to flood *DHCP Discover* messages with unique *CHADDRs* (*Client Hardware Address*, which indicates the MAC address of the client).
- This leads to the *target's* *DHCP pool* to become full, resulting in a *DoS*.

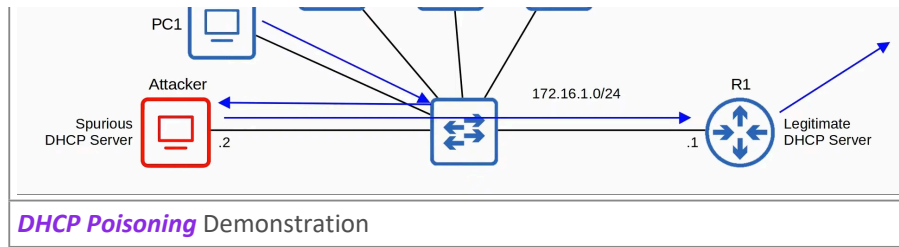


2. DHCP Poisoning (Man-in-the-middle)

DHCP Poisoning is similar to *ARP Poisoning*. It can be used to perform a man-in-the-middle attack.

- A *spurious* (*malicious*) *DHCP* server replies to clients' *DHCP Discover* messages and assigns them IP addresses, but makes the client use the *spurious* server's IP as the default gateway.
- (*Clients usually accept the first DHCP Offer message they receive*)
- This will cause the client to send traffic to the *attacker* instead of the legitimate default gateway of the network.
- The *attacker* can then examine/modify the traffic before forwarding it to the legitimate default gateway.





DHCP Message Types

There are two types of DHCP messages, DHCP **Server** messages and DHCP **Client** messages.

1. **DHCP Server**
 - *Offer*
 - *ACK*
 - *NAK* (Opposite of *ACK*, used to decline a client's **DHCP Request**)
2. **DHCP Client**
 - *Discover*
 - *Request*
 - *Release* (Used to release a client's IP address)
 - *Decline* (Used to decline the IP address offered by a **DHCP** server)

DHCP snooping has the ability to differentiate and filter between both of them.

DHCP Snooping - Configurations

- `show ip dhcp snooping binding`
 - Shows the **DHCP snooping** binding table
 - Includes information such as:
 - *MAC Address*
 - *IP Address*
 - *Lease time*
 - *Type*
 - *VLAN*
 - *Interface*
- `ip dhcp snooping`
 - Enables the **DHCP snooping** feature globally to be used on the switch.
- `ip dhcp snooping vlan VLAN-ID`
 - Enables **DHCP snooping** on a specific VLAN.
- `ip dhcp snooping trust`
 - Configures the current interface as a trusted **DHCP snooping** interface.
- `ip dhcp snooping limit rate PACKETS-PER-SECOND` ^dhcp-snooping-limit-rate
 - Limits how many **DHCP** packets are allowed per second.
- `no ip dhcp snooping information option`
 - Disables applying **Option 82** for **DHCP** messages.
 - ([What is Information Option 82?](#))
- `errdisable recovery cause dhcp-rate-limit`
 - Enables err-disable recovery for **DHCP snooping** rate limits.

DHCP Snooping - Operations

- When a client successfully leases an IP address from the **DHCP** server, create a new entry in the **DHCP snooping binding table**.
- If a **DHCP** message is received on a *trusted* port, forward it as normal without inspection.
- If a **DHCP** message is received on an *untrusted* port, examine it and act as follows:
 1. If it's a **DHCP Server** message:
 - Discard the message.
 2. If it's a **DHCP Client** message, perform the following checks:

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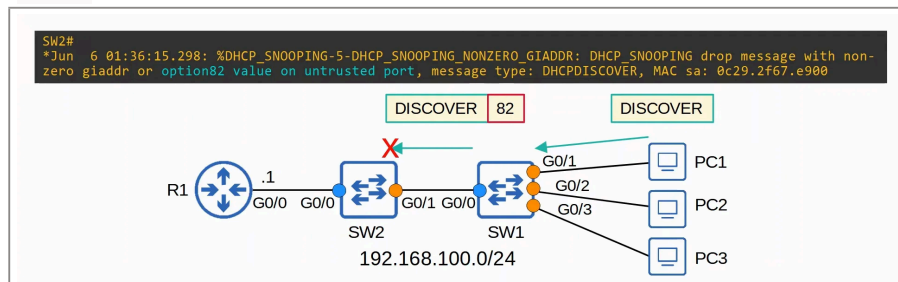
- **Discover/Request:**
 - Check if the frame's **source** MAC address and **DHCP** message **CHADDR** fields match or not.
 - Match = **Forward**
 - Mismatch = **Discard**
- **Release/Decline:**
 - Check if the packet's **source** IP address and the **receiving** interface match the entry in the **DHCP snooping binding table**.
 - Match = **Forward**
 - Mismatch = **Discard**

DHCP Snooping - Rate-Limiting

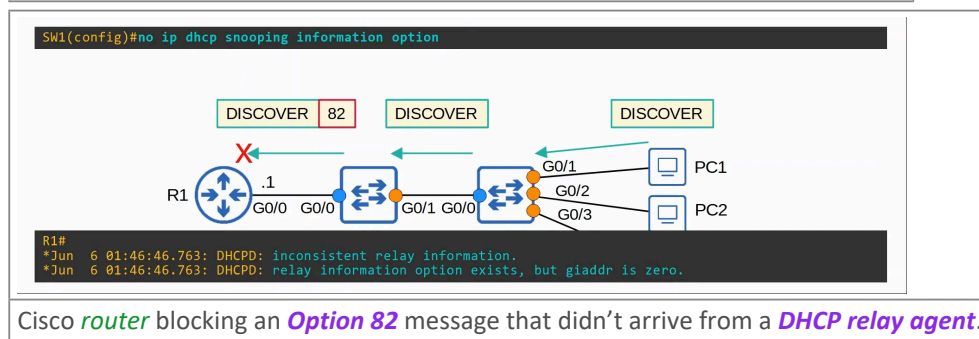
- Limits how many **DHCP** packets are allowed per second.
- **DHCP Rate-limiting** is disabled on all interfaces by default.
- If the rate of **DHCP** messages goes over the configured limit, the interface goes into **err-disabled** mode.
 - The rate can be configured through the `ip dhcp snooping limit rate PACKETS-PER-SECOND` command.
- Can be re-enabled with `shutdown` ⇒ `no shutdown`, or through ErrDisable Recovery (`errdisable recovery cause dhcp-rate-limit`).

DHCP Snooping - Option 82 (Information Option)

- **Option 82**, also known as the '**DHCP relay agent information option**' is one of many **DHCP** options. It is usually used when there's a remote **DHCP** server.
- **DHCP relay agents** can add **Option 82** to messages they forward to the remote **DHCP** server.
- It provides additional information about which **DHCP relay agent** received the client's message, on which interface, in which VLAN, and etc.
- It can be disabled with `no ip dhcp snooping information option`
- When **DHCP snooping** is enabled, by default Cisco switches will add **Option 82** to **DHCP** messages they receive from clients, ==even if the switch isn't acting as a **DHCP relay agent**==.
- By default, Cisco switches will drop **DHCP** messages with **Option 82** that are received on an **untrusted** interface.
- Also by default, Cisco routers will drop **DHCP** messages with **Option 82** that haven't been sent by a **DHCP relay agent**.



DHCP snooping on a Cisco **switch** blocking an **Option 82** message.



Cisco **router** blocking an **Option 82** message that didn't arrive from a **DHCP relay agent**.